

A latent process model for discrete extremes

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Outline of the talk

- 1 Motivation
- 2 A distribution for discrete extreme values
- 3 Time series models for discrete extreme values
- 4 Inference and numerical examples
- 5 Concluding remarks

Motivation

- Extreme events could occur in clusters; for instance,
 - Rainfall: Extreme rainfall over the catchments
 - Floods/hurricane: Extreme events extend over a region w.r.t to time.
 - Seasonal flu: Occurs in clusters/ Waves, e.g., Influenza and COVID-19?

Motivation

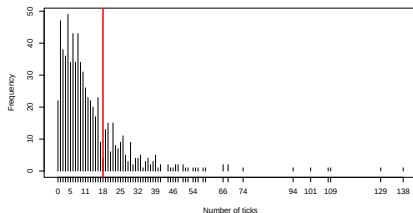
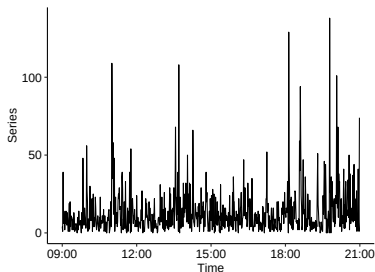
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- Our motivation is to model
 - **integer valued times series** associated with extreme events (e.g., number of tick changes by minutes of the exchange rate, number of insurance claims, etc.)
 - possible **temporal dependence** in the tail.

Extremes of exchanges rates (Motivating example)

- Number of tick changes by minutes of the exchange rate of euro to British pound (EUR/GBP) on December 12th, 2019 (Gorgi, 2020).



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- Generalized Pareto distribution (GPD)

$$F_{\text{GPD}}(x; \sigma, \xi) = \begin{cases} 1 - (1 + \xi x / \sigma)^{-1/\xi}, & \xi \neq 0 \\ 1 - e^{(-x/\sigma)}, & \xi = 0 \end{cases} \quad (1)$$

approximates the distribution of exceedances over the high threshold.

A distribution for discrete extreme values

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- Shorthand: $Y \sim \text{DGPD}(\xi, \sigma)$

A useful representation of DGPD

Hierarchical model

Stage 1: $Y|\Lambda \sim \text{Geo}(p)$, $p = 1 - e^{-\Lambda/\beta}$, $\beta > 0$

Stage 2: $\Lambda \sim \text{Gamma}(\alpha, 1)$, $\alpha > 0$

then

$$Y \sim \text{DGPD}(\xi, \sigma), \quad \xi = 1/\alpha, \sigma = \beta/\alpha$$

- **Why hierarchical model?**

Hierarchical models are flexible to combine different dependence layers with observed data:

Time series models for discrete extreme values

Basic idea

- Suggest a model for the time series $\{Y_t, t = 1, 2, \dots\}$ when Y_t is greater of a threshold u
- Tail distribution of Y_t consistent with a **DGPD distribution**.

Hierarchical model for time series

Stage 1: $Y|\Lambda_t \sim \text{Geo}(p)$, $p = 1 - e^{-\Lambda_t/\beta}$, $\beta > 0$

Stage 2: $\{\Lambda_t\}$ is a **latent random process** with $\text{Gamma}(\alpha, 1)$ marginals

then

$$Y_t \sim \text{DGPD}(\xi, \sigma), \quad \xi = 1/\alpha, \sigma = \beta/\alpha$$

- The **latent random process** induces dependence.

Pairwise distributions

- Marginal and pairwise distributions play an important role in
 - extremal properties
 - inference

$$\Pr(Y_t > k) = \mathbb{E} \left(e^{-\frac{k\Lambda_t}{\beta}} \right) = L_t^{(1)}(k/\beta)$$

$$\Pr(Y_{t_1} > k_1, Y_{t_2} > k_2) = \mathbb{E} \left(e^{-\frac{k_1\Lambda_{t_1}}{\beta} - \frac{k_2\Lambda_{t_2}}{\beta}} \right) = L_{t_1, t_2}^{(2)}(k_1/\beta, k_2/\beta)$$

- $L_t^{(1)}(\cdot)$ Laplace transforms of r.v. Λ_t
- $L_{t_1, t_2}^{(2)}(\cdot, \cdot)$ Laplace transforms of bivariate r.v. $(\Lambda_{t_1}, \Lambda_{t_2})$

Tail dependence measures I

- Suppose that $\{\Lambda_t\}$ is stationary $\Rightarrow \{Y_t\}$ is stationary

Upper tail dependence coefficient for (Y_t, Y_{t+j})

$$\chi_j := \lim_{k \rightarrow \infty} \frac{\Pr(Y_t > k, Y_{t+j} > k)}{\Pr(Y_t > k)} = \lim_{k \rightarrow \infty} \frac{L_{t,t+j}^{(2)}(k, k)}{L_t^{(1)}(k)}$$

- Asymptotic dependence arises when $\chi_j > 0$
- Asymptotic independence corresponds to $\chi_j = 0$.

Tail dependence measures II

In case of asymptotic independence ($\chi_j = 0$)

coefficient of residual tail dependence for (Y_t, Y_{t+j})

$$\eta_j := \lim_{k \rightarrow \infty} \frac{2 \log \Pr(Y_t > k)}{\log \Pr(Y_t > k, Y_{t+j} > k)} - 1 = \lim_{k \rightarrow \infty} \frac{2 \log L_t^{(1)}(k)}{\log L_{t,t+j}^{(2)}(k, k)} - 1$$

Which latent random process ? I

Gaver and Lewis process (Gaver and Lewis, 1980)

$$\begin{aligned}\Lambda_t &= \rho\Lambda_{t-1} + V_t, & 0 \leq \rho < 1 \\ \Lambda_{t-1} &\sim \text{Gamma}(\alpha, \beta)\end{aligned}$$

for iid $\{V_t\}$ with CHF $E(e^{i\omega V_t}) = \left[\frac{\beta - i\omega}{\beta - i\rho\omega} \right]^{-\alpha}$ (Wolpert, 2021).

- $\text{Corr}(\Lambda_t, \Lambda_{t+j}) = \rho^j$
- $\chi_j = \left[\frac{\rho^j}{1+\rho^j} \right]^\alpha > 0, \Rightarrow \text{asymptotic dependence}$

Which latent random process ? II

Thinned Gamma process (Wolpert, 2021)

$$\begin{aligned}\Lambda_t &= B_t \Lambda_{t-1} + V_t, & 0 \leq \rho < 1 \\ \Lambda_{t-1} &\sim \text{Gamma}(\alpha, \beta)\end{aligned}$$

for iid $\{B_t\}$, $B_t \sim \text{Beta}(\alpha\rho, \alpha(1-\rho))$ and iid $\{V_t\}$,
 $V_t \sim \text{Gamma}(\alpha(1-\rho), \beta)$

- $\text{Corr}(\Lambda_t, \Lambda_{t+j}) = \rho^j$
- $\chi_j = 0, \Rightarrow$ asymptotic independence
- $\eta_j = \frac{1}{2-\rho^j}$

Inference and numerical examples

Pairwise likelihood inference

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- We have observations censored under a high threshold u
- Full likelihood is too cumbersome but the pairwise likelihood is computationally attractive
- We estimate the parameters $\theta = (\alpha, \beta, \rho)$ through pairwise likelihood (Lindsay, 1988).

$$pl_n(\theta) = \sum_{i=1}^{n-1} \sum_{j=i+1}^{\min(i+\Delta, n)} L(y_i, y_j; \theta) \quad (2)$$

- $L(k_i, k_j)$ is the log-likelihood of the pair of (eventually censored) observations (y_i, y_j)
- $1 \leq \Delta \leq n - 1$ is a value which defines the maximum lag.

Simulation study

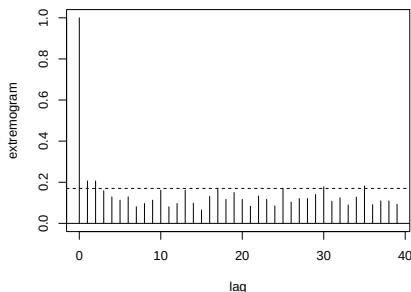
Table 1: Root mean squared error and bias for α and ρ found from simulations from the models based on GLP and TGP, with $n=30,000, (u = 0.90)$ quantile and under different parameter settings.

lag	Tv	α		Tv	ρ	
		GLP	TGP		GLP	TGP
1	5	0.27	0.28	0.40	0.05	0.06
		(0.002)	(0.02)		(0.00)	(-0.003)
	5	0.32	0.30	0.90	0.06	0.04
		(0.02)	(0.02)		(-0.002)	(0.00)
	10	0.97	0.97	0.40	0.10	0.10
		(0.09)	(0.11)		(0.007)	(-0.001)
7	10	1.01	0.98	0.95	0.09	0.07
		(0.04)	(0.02)		(-0.03)	(-0.02)
	5	0.27	0.26	0.40	0.04	0.04
		(0.01)	(0.01)		(-0.001)	(-0.005)
	5	0.32	0.30	0.90	0.01	0.01
		(0.02)	(0.01)		(-0.001)	(-0.002)
	10	0.97	0.97	0.40	0.07	0.08
		(0.09)	(0.11)		(0.000)	(-0.001)
	10	1.02	1.03	0.95	0.02	0.02
		(0.11)	(0.10)		(-0.001)	(-0.001)

Inside the table, the Tv, GLP and TGP stand for True values, Gaver and Lewis process, and Thinned gamma process, respectively.

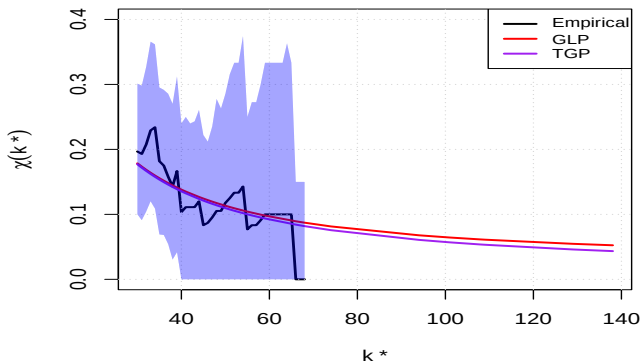
Real data example

- Number of tick changes by minutes of the exchange rate of euro to British pound (EUR/GBP) on December 12th, 2019.



Extremal correlation

- GLP, TGP and other models are fitted to the number of tick changes by minute of EUR/GBP sterling exchange rate time series.
- Threshold $u = 29$, lag $\Delta = 4$.



Concluding remarks

- The proposed model has demonstrated good flexibility in capturing different forms of extremal dependence.
- We can achieve asymptotic clustering and asymptotic independence of extremes by choosing an appropriate latent process Λ_t .

⇒ Ongoing research

- Multivariate modeling on graph
- Space-time modeling

Thank you for your attention!

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